

Business Problem Statement

Retail Customer Intelligence & Revenue Optimization

Business Context

A retail business wants to better understand customer purchasing behavior in order to improve customer engagement, product positioning and promotion strategy. Management needs a clear view of which categories and products contribute the most purchase value, how customer behavior differs across segments, and how subscription, discount, shipping and payment behavior are associated with purchases.

Business Problem

The company has customer-level shopping data, but needs to convert that data into actionable business insights. The central business question for this project is:

“How can the business use customer shopping data to understand purchasing behavior, optimize product and promotion strategies, and improve customer engagement?”

Key Business Questions

1. What is the overall purchase value and average purchase amount?
2. Which categories contribute the most purchase value?
3. Which products are the strongest contributors to purchase value?
4. Which categories and products receive stronger customer ratings?
5. How does purchase behavior differ between discounted and non-discounted transactions?
6. Which categories have greater discount usage?
7. How does subscription status relate to average purchase behavior?
8. Which purchase-frequency groups show higher average purchase amounts?
9. Which payment and shipping options are most commonly used?
10. Which customer behavior patterns can support better segmentation?

Project Deliverables

11. Data Preparation & Modeling (Python): Clean, validate and transform the raw customer dataset.
12. Business Analysis (PostgreSQL): Load the cleaned dataset and answer the defined business questions using SQL.
13. Visualization & Insights (Power BI): Build an interactive three-page dashboard for executive, customer/promotion and product/experience analysis.
14. Business Recommendations: Translate analytical findings into practical actions and clearly state dataset limitations.
15. GitHub Portfolio: Organize the notebook, SQL queries, cleaned data, Power BI dashboard and documentation in a professional repository.

Expected Business Outcomes

- Identify high-value product categories and products.
- Understand customer purchasing and purchase-frequency patterns.
- Identify useful customer segmentation variables.
- Understand how subscription and promotion behavior differs across customers.
- Highlight product and category areas associated with stronger customer ratings.
- Provide practical recommendations for cross-selling, customer engagement and future promotion evaluation.

Analytical Scope & Constraints

- No transaction date; monthly, yearly, cohort and retention trends are not included.
- No profit or margin field; profitability and promotion ROI are not evaluated.
- The analysis identifies patterns and associations rather than proving causal relationships.